Exam #2 (Sta-209, S25)

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Formula Sheet

Definitions:

- Type I error - rejection of a true null hypothesis
- Type II error - failure to reject a false null hypothesis

Central Limit theorem results:

Statistic	Standard Error	Conditions
$\hat{p}$	$\sqrt{\frac{p(1-p)}{n}}$	$np \geq 10$ and $n(1-p) \geq 10$
$\bar{x}$	$\frac{\sigma}{\sqrt{n}}$	normal population or $n \geq 30$
$\hat{p}_1 - \hat{p}_2$	$\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$	$n_i p_i \geq 10$ and $n_i(1-p_i) \geq 10$ for $i \in \{1,2\}$
$\bar{x}_1 - \bar{x}_2$	$\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$	normal populations or $n_1 \geq 30$ and $n_2 \geq 30$

Question #1 (conceptual questions)

Part A: Recall that a p-value is a probability. In your own words, briefly explain what the p-value describes the probability of. Limit your answer to at most 2-sentences.

Part B: Explain the meaning of "95% confidence" in the statistical term *95% confidence interval*. Do not use the words "confident" or "confidence" as core components of your explanation. Limit your answer to at most 2-sentences.

Part C: A team of researchers perform a hypothesis test comparing the survival rate of two groups and find a p-value of 0.00001. Assuming there are no flaws in the experimental design, and no confounding that wasn't controlled for, are you confident that these researchers have made an *extremely important scientific finding*? Answer either "yes" or "no" and explain your reasoning using at most 2-sentences.

Part D: A researcher is trying to decide whether to use 95% confidence intervals or 99% confidence intervals in their analysis. What is *one advantage* and *one disadvantage* of using the 99% confidence level (rather than the 95% confidence level)? Be sure to state both an advantage and a disadvantage.

Part E: In a genomics experiment involving 20,000 hypothesis tests, a statistician uses a procedure that controls the *false discovery rate* at 5% and identifies 80 significant genes. If they had instead used the Bonferroni Correction to compare each of their original p-values to an adjusted threshold of $\alpha^* = 0.05/20000$, or 0.0000025, would you expect them to identify more than 80 significant genes, exactly 80 significant genes, or fewer than 80 significant genes? Briefly explain your answer, limiting your response to at most 3-sentences.

Question #2

In a 1997 research paper, psychologists Ruback and Juieng performed a series of studies investigating how quickly a person leaves their parking spot.

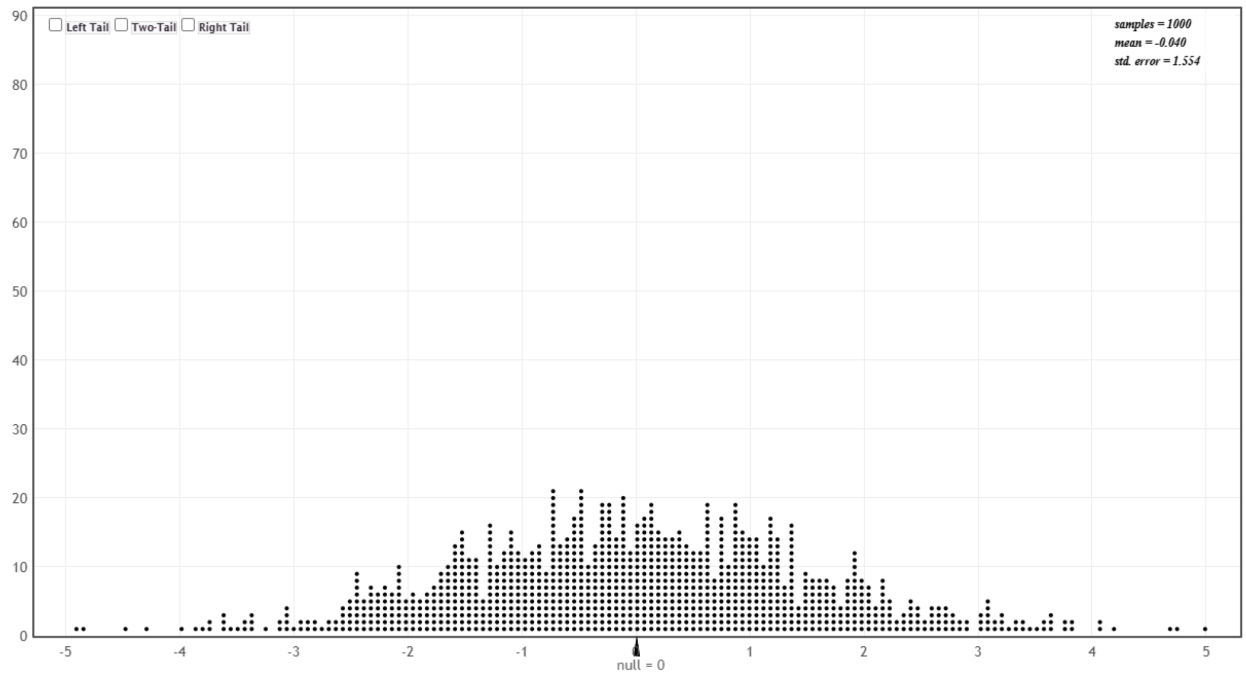
In one of these studies, Ruback and Juieng observed 200 drivers departing from a public parking lot. For each departing driver they recorded the time (in seconds) from when the driver first entered their car to when their vehicle had fully exited their parking space. The main explanatory variable in the study was whether or not another car was waiting for the driver's parking space while they were exiting the parking lot, and the researchers hypothesized that people would exit more quickly when someone was waiting for their spot.

```
## Descriptive Statistics
parking %>%
  group_by(waiting) %>%
  summarize(mean(time), median(time), sd(time), IQR(time), n())
```

```
## # A tibble: 2 x 6
##   waiting `mean(time)` `median(time)` `sd(time)` `IQR(time)` `n()`
##   <chr>          <dbl>          <dbl>      <dbl>       <dbl> <int>
## 1 no              45.5           46.1      10.5        13.9    132
## 2 yes             49.4           48.1       9.42        9.57    68
```

Part A: State the null and alternative hypotheses in this application using *both* words and statistical notation.

Part B: Shown below are 1000 outcomes that were simulated under the null hypothesis you proposed in Part A. Use this distribution to estimate the two-sided p-value. You may choose to annotate the figure to indicate to me how you are estimating the p-value.

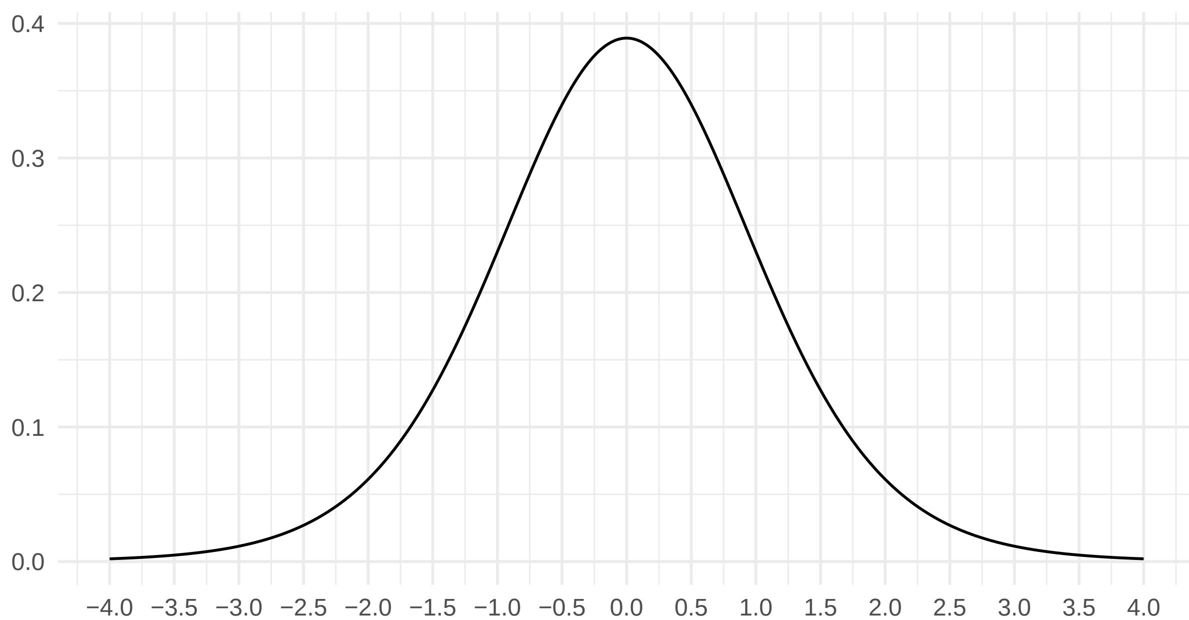


Part C: Provide a 1-sentence summary of this study based upon the p-value you estimated in Part B.

Part D: Is it possible that the conclusion you made in Part C is a Type II error? Briefly explain.

Part E: Suppose you'd like to use a statistical test based upon a probability model rather than simulation to evaluate the hypotheses from Part A. The test statistic in this scenario uses the standard error formula $SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$. Calculate this test statistic. Show your work.

Part F: Shown below is the probability model (t-distribution) for the test statistic you calculated in Part E. Shade the area of this distribution that represents the *two-sided p-value* in this application.



Question #3

At many colleges, including Grinnell, when a faculty member is undergoing review, the Dean's Office will contact a sample of students who have taken a course with the faculty member and ask them several survey questions. These students are selected at random. Suppose one question asked during the survey is whether the student found the faculty member's course to be worthwhile in helping them achieve their academic and career goals. Students have the option to answer "yes" or "no" to this question.

Part A: Suppose that for a certain faculty member's review the Dean's Office contacts a random sample of 25 students, and all of them respond to the survey, with 18 answering "yes" to the aforementioned question. Use this information to calculate a 95% confidence interval estimate for the proportion of all former students who found the faculty member's class worthwhile. You should use the calibration constant $c = 1.96$ to achieve the stated confidence level. Show your work.

Part B: Based upon the interval you calculated in Part A, do you believe it possible that students were responding randomly to the aforementioned question? Briefly explain.

Part C: Assuming everything else remains constant, which of the following would *decrease* the width of the confidence interval estimate you calculated in Part A? Clearly indicate *all* of the options that would result in narrower interval. You do not need to explain your answers.

i) The Dean's Office contacted 50 former students rather than 25.
ii) You calculated a 90% confidence interval rather than a 95% confidence interval
iii) The Dean's Office continued to recruit former students until the interval had a margin of error of 0.05
iv) You incorrectly chose your value of "c" to reflect the middle 95% of the t-distribution rather than the Standard Normal distribution

Part D: Suppose the Dean's Office is reviewing a different faculty member and they contact 40 former students who were randomly selected, but only 25 of the contacted students respond. Explain why *sampling bias* might be a concern in this scenario.

Part E: Suppose that next year the Dean's Office plans review 30 different faculty members, and for each of these faculty they will calculate a 95% confidence interval estimate for the proportion who answer "yes" to the aforementioned question. Assuming every interval is made using a valid procedure, would you expect more of these intervals to contain the true proportion of former students who believe the faculty benefited them if the Dean's office decides to gather responses from 50 students rather than 25 students? Explain your reasoning.